RAMCO INSTITUTE OF TECHNOLOGY
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Department of Information Technology
Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech IT
Course Code & Title : CS3451 & Introduction to Operating Systems
Name of the Faculty member : Mr.S. Sakkaravarthi, AP/IT
Mr. Balaganesh M V, AP/IT

Innovative Practice Description

- **Unit / Topic: Unit IV / Organization of Records in Files**
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO 8
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Flipping the classroom is an inverting the classroom approach to teaching. In this approach, the traditional in-class teaching is "flipped" to better meet the needs of individual learners. Students gain control of the learning process through studying course material outside of class, using readings, pre-recorded video lectures. It helps the faculty/lecturer to redefine in-class activities and include homework problems and keep the students engaged in the content.

- **Time Allotted for the Activity:** 20 Minutes
- **Details of the Implementation:**
  - **Plan:** Basics of organization of records in files are already learnt in the previous semester and few topics related to DBMS is given as self-learning through self-exploration.
  - **Identify and Share**: Related materials are identified and I posted relevant references/materials/notes of File Organization mechanisms to the students and given as self-learning. The Figure.1 shows the activities / discussion as a team on the self-exploration topics.
  - **Evaluate:**
    - ➢ I have prepared few questions related to the content shared and ask those questions in next class session and make the students to write/present in class. The Figure.1 images shows the student Nandhini S and Ananthi P explaining the self-explored concepts on File Organizations.

➢ Make the students to find out the answer by their own by learning.

CO	PO1	PO4	PO9	PO10	PSO1
CO4	**2**	**2**	**1**	**1**	**1**

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO4**	**PO9**	**PO10**	**PSO2**
	2	**2**	**1**	**1**	**1**
Justification for correlation	Students could apply the disk scheduling algorithms to analyze the performance in terms of disk arm movement and delay.	Students will be able to conclude which disk scheduling can be followed to improve the throughput of disk response.	To work as an individual, and as a member	To Communicate effectively on complex engineering activities	The basic understanding of the disk scheduling algorithms will make the students to excel in choosing the suitable disk scheduling algorithms in designing Operating systems

- **Images / Screenshot of the practice:**


Fig 1 & Fig 2. Nandhini S (Left) and Ananthi P (Right) explaining the concepts in the class.

- **Reflective Critique:**
  - ❖ ***Feedback of practice from students and other stakeholders:***
    - Student felt good, since they can study at their own pace/time.
    - They felt that through such learning, they can explore more.

❖ ***Benefit of the practice:***

- More one-to-one time between teacher and student.
- More collaboration time for students.
- Students learn at their own pace.
- Practical things – like missing class due to illness – become less problematic.
- It encourages students to come to class prepared.

❖ ***Challenges faced in implementation:***

- Relies on student preparation – few students did not come prepared.
- The depth of the subject can be dictated by the student themselves or the group the student is working with.
- The time and effort required from a teacher's perspective initially when creating the flipped class material is higher than for a traditional class

References:

❖ https://www.teachthought.com/learning/the-definition-of-the-flipped-classroom/#:~:text=A%20flipped%20classroom%20is%20a,the%20students%20i

Signature of Faculty Member **HOD**